specific about the safe work procedures needed for your workplace. (For example, the SDS may not specify what type of respirator must be used, just that a respirator is needed.) Ask your supervisor for more information. These decisions may require the help of a safety professional or someone with chemical safety knowledge.

MSDS headings vs. SDS headings

The table below compares the section headings of a sample MSDS against the headings of an SDS. Note that there was no set MSDS format, but the SDS format is standardized. The MSDS headings shown below are taken from a form on worksafebc.com.

Table 7. MSDS headings vs. SDS headings

9 section MSDS headings	16 section WHMIS 2015 SDS headings
1. Product information	1. Identification
2. Hazardous ingredients	2. Hazard identification
3. Physical data	3. Composition/Information on ingredients
4. Fire and explosion data	4. First-aid measures
5. Reactivity data	5. Fire-fighting measures
6. Toxicological properties	6. Accidental release measures
7. Preventive measures	7. Handling and storage
8. First aid measures	8. Exposure controls/Personal protection
9. Preparation information	9. Physical and chemical properties
	10. Stability and reactivity
	11. Toxicological information
	12. Ecological information*
	13. Disposal considerations*
	14. Transport information*
	15. Regulatory information*
	16. Other information

* In sections 12 to 15, the headings need to be listed, but suppliers don't have to provide information.

Looking beyond an SDS for more information

An SDS may not contain all the information you need. A lot of health hazard information, for example, is written in general terms. And SDSs are often written for many different uses of products. So the handling and safety precautions may not be specific to your workplace.

The following people or organizations should be able to help you find more information if needed:

- Your health and safety committee or representative
- A health and safety specialist
- An occupational health nurse
- Your family doctor
- Your supervisor
- Your employer
- Suppliers

Confidential business information

Confidential business information (CBI) refers to specific product information that suppliers are permitted to withhold from an SDS or label for a period of three years. Under WHMIS, a supplier can make a request to Health Canada to protect certain information that gives a company a business advantage over competitors. Crucial information such as health hazards may never be withheld.

Here's a quick summary of how CBI works:

- Suppliers may apply for confidential business information protection to Health Canada.
- An approved claim is valid for three years.
- Protected trade information is only released to health or safety professionals in case of an emergency.
- Health hazard information must be shown on the SDS.

6 Putting WHMIS into action

WHMIS program

What is an exposure control plan?

An exposure control plan (ECP) sets out a detailed approach to protecting workers from harmful exposure to certain substances (for example, carcinogens) and, under some conditions, for all other hazardous products.

If your workplace uses hazardous products, a WHMIS program must be in place. To put in place a WHMIS program, your employer needs to make use of supplier labels and SDSs.

Your employer also needs to use his or her own knowledge of the hazards of products and their use in the workplace. This knowledge should take into account factors such as work processes, control measures, and work location. For example, the hazards of spray painting with a hazardous product inside a confined space are far different from the hazards of hand brushing the same product outdoors.

Based on all of this information, your employer must develop exposure control plans when required and written safe work procedures that ensure your health and safety. Your employer must also educate you about the hazards and train you in safe work procedures.

Your health and safety committee or representative must be involved in the development, implementation, and review of the WHMIS program.

Employer responsibilities for a WHMIS program

- Assign responsibility.
- Establish an inventory of hazardous products.
- Meet SDS and label requirements.
- Determine the hazards of hazardous products.
- Establish workplace controls.
- Establish emergency procedures.
- Provide worker education and training.
- Evaluate the WHMIS program.

Education and training

Your employer is responsible for educating you about WHMIS and training you in safe work procedures.

Who should receive education and training?

As a worker, you must be educated and trained so you understand the hazards and know how to work safely with hazardous products.

If you work with a hazardous product, or may be exposed to a hazardous product as part of your work activities, you must learn about the hazard information for that product. (For example, a receptionist at a medical clinic may be exposed to mercury if it is spilled.) The hazard information should include the information received from the supplier. It should also include any other information that your employer is aware of about the use, storage, and handling of each product.

For instance, you will receive this education and training if one or more of the following applies:

- You may be exposed to a hazardous product due to your work activities (including normal use, maintenance activities, or emergencies).
- You use, store, handle, or dispose of a hazardous product.
- You supervise or manage other workers who may be exposed, or use, store, handle, or dispose of a hazardous product.
- You are involved in emergency response.

Who should provide the education and training?

Under WHMIS, your employer is responsible for education and training. WHMIS outlines those minimum requirements. Your employer may provide the education and training, or it may be provided by a qualified person or agency chosen by your employer. Regardless of who delivers the education and training, employers remain legally responsible to ensure the protection of workers.

Topics to be covered

Examples of topics that should be covered during education and training include the following:

- The information on both supplier labels and workplace labels, and what that information means
- The information on the safety data sheet (SDS) and what that information means
- The procedures required for safe use, handling, and disposal of a hazardous product
- Any other information required when the product is in a pipe, piping system, vessel, tank car, etc.
- Procedures to follow if the hazardous product may be present in the air and you may be exposed
- All procedures that you must follow in an emergency that involves the hazardous product

Your responsibilities

As a worker, you must participate in the education and training sessions. You must also follow your employer's safe work procedures.

If you have been successfully educated and trained in WHMIS, you must be able to answer these four questions for every hazardous product you work with:

- What are the hazards of the product?
- How do I protect myself from those hazards?
- What do I do in case of an emergency?
- Where can I get more information?

7 Exercises

Pictogram matching exercise

Match the pictograms to the hazards.

Pictogram		Hazard
1.		**A.** Organisms or toxins that can cause diseases in people or animals
2.		**B.** May cause or suspected of causing serious health effects
3.		**C.** Oxidizing hazard
4.		**D.** Gases under pressure
5.		**E.** Fire hazard
6.		**F.** Explosion or reactivity hazard
7.		**G.** May cause less serious health effects or damage the ozone layer
8.		**H.** Can cause death or toxicity with short exposure to small amounts
9.		**I.** Corrosive damage to metals, as well as skin, eyes
10.		**J.** May cause damage to the aquatic environment

Label exercise

1. What is the purpose of a WHMIS label?

_____

2. What are the two main types of WHMIS labels?

_____

3. Identify the types of information on a WHMIS 2015 supplier label from your workplace. (Most supplier labels will show six types of information, but some may show seven.)

- _____
- _____
- _____
- _____
- _____
- _____
- _____

4. Identify the three types of information on this workplace label.

Solv-easy

Keep away from sparks, heat, and open flame.

Use local exhaust ventilation or NIOSH-approved organic vapour respirator.

Wear neoprene gloves and chemical splash goggles.

See the SDS.

5. Name two situations where other means of identification can be used.

- _____
- _____

SDS exercise

1. Where are the SDSs kept in your workplace?

_____

_____

2. With some exceptions, employers need to make sure that no SDS is more than _____ years old.

3. How many sections are there on an SDS?

_____

_____

4. What are five key things that an SDS tells you?

_____

_____

_____

_____

_____

5. Using an SDS for a product in your workplace, review the hazards of the product, the safe handling procedures, personal protective equipment, and storage and shipping requirements for the product. Write your key findings in the space below.

_____

_____

_____

_____

_____

8 Answer keys

Pictogram matching exercise answers

Match the pictograms to the hazards.

Pictogram		Hazard	Answer
1.		**A.** Organisms or toxins that can cause diseases in people or animals	**1-G**
2.		**B.** May cause or suspected of causing serious health effects	**2-H**
3.		**C.** Oxidizing hazard	**3-F**
4.		**D.** Gases under pressure	**4-B**
5.		**E.** Fire hazard	**5-C**
6.		**F.** Explosion or reactivity hazard	**6-A**
7.		**G.** May cause less serious health effects or damage the ozone layer	**7-D**
8.		**H.** Can cause death or toxicity with short exposure to small amounts	**8-E**
9.		**I.** Corrosive damage to metals, as well as skin, eyes	**9-J**
10.		**J.** May cause damage to the aquatic environment	**10-I**

Label exercise answers

1. What is the purpose of a WHMIS label?
   Answer: Labels are the first alert to users about the major hazards of hazardous products. They also outline the basic precautions or safety steps that should be taken.

2. What are the two main types of WHMIS labels?
   Answer: Supplier labels and workplace labels

3. Identify the types of information on a WHMIS 2015 supplier label from your workplace. (Most supplier labels will show six types of information, but some may show seven.)
   Answers:
   - Product identifier
   - Initial supplier identifier
   - Pictogram(s)
   - Signal word
   - Hazard statement(s)
   - Precautionary statement(s)
   - Supplemental label information (in rare cases)

4. Identify the three types of information on this workplace label.

Solv-easy

Keep away from sparks, heat, and open flame.

Use local exhaust ventilation or NIOSH-approved organic vapour respirator.

Wear neoprene gloves and chemical splash goggles.

See the SDS.

   Answers: Product identifier, safe handling/precautionary information, and reference to safety data sheet

5. Name two situations where other means of identification can be used.
   Answer: Two of the following:
   - Bulk shipments
   - Small capacity containers, 100 mL or less
   - Small capacity containers, 3 mL or less
   - Piping systems and vessels
   - Laboratory samples

SDS exercise answers

1. Where are the SDSs kept in your workplace?
   Answer: Identify location. (Note that the location must provide easy access to SDSs at all times, whether in hard copy or electronic format.)

2. With some exceptions, employers need to make sure that no SDS is more than _____ years old.
   Answer: Three

3. How many sections are there on an SDS?
   Answer: 16 (Note that suppliers have the option not to provide information for sections 12 to 15.)

4. What are five key things that an SDS tells you?
   Answer:
   - The hazards of a product
   - How to use the product safely
   - What to expect if you don't follow the advice
   - How to recognize symptoms of exposure
   - What to do if emergencies occur

5. Using an SDS for a product in your workplace, review the hazards of the product, the safe handling procedures, personal protective equipment, and storage and shipping requirements for the product. Write your key findings in the space below.
   Answer: Review and discuss your findings.

9 Appendix

Information items on a supplier label

If you want to check whether a supplier label shows all the components required, use the following template. A similar template for checking SDSs is available in the WorkSafeBC publication *WHMIS 2015: At Work*, which can be found at worksafebc.com.

Information item		Description
1	Product identifier	The brand name, chemical name, common name, generic name, or trade name of the hazardous product.
2	Initial supplier identifier	The name, address, and telephone number of either the Canadian manufacturer or the Canadian importer.
3	Pictogram(s)	A hazard symbol usually contained within a red, diamond-shaped border.
4	Signal word	One of just two words used to alert you to a potential hazard and to state the severity of the hazard. "Danger" is used for high-risk hazards, while "Warning" is used for less severe hazards.
5	Hazard statement(s)	A standardized phrase or phrases that describe the type of hazard(s) posed by a hazardous product. Examples include "Extremely flammable gas," "Fatal if inhaled," and "May cause cancer."
6	Precautionary statement(s)	A standardized phrase or phrases that describe how to reduce or prevent harmful effects resulting from the following: • Exposure to a hazardous product, or • Improper handling or storage of a hazardous product Examples include "Keep container tightly closed," "Wear eye protection," and "If exposed or concerned: Get medical attention."

In rare cases, labels may include supplemental label information about the following:

- Precautionary actions
- Hazards not yet included in GHS
- Physical state (gas, liquid, solid, etc.)
- Route of exposure (by inhaling, by swallowing, through skin, etc.)

Supplemental label information is only included on a supplier label in the following cases:

- A toxic mixture has an ingredient with unknown acute toxicity, or
- A product reacts with water to produce an acutely toxic gas

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